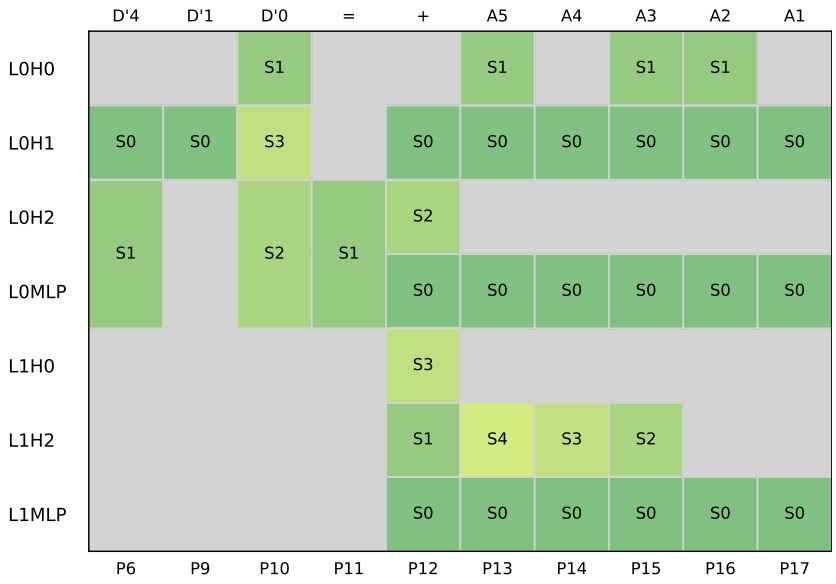

Figure 13: This map shows the simpliest (lowest **complexity**) quanta S0, S1, etc impacted when we ablate each node in the **5-digit** 2-layer 3-head **addition** model. To answer S0 questions, only the S0 nodes are used. To answer S1 questions, S0 and S1 nodes are used, etc.

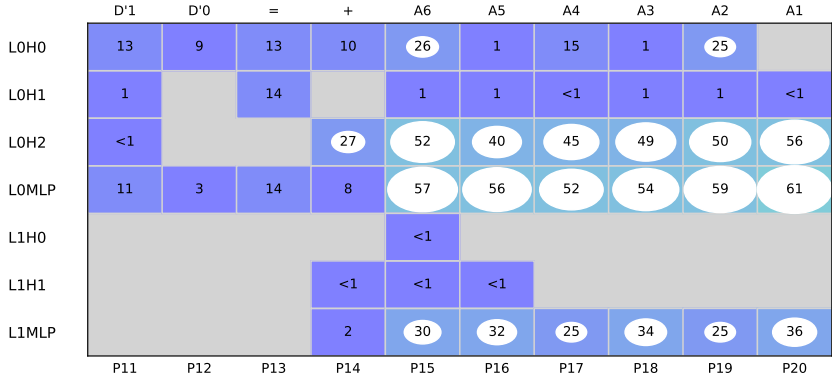

Figure 14: This map shows the **% of questions** that fail when we ablate each node in the **6-digit** 2-layer 3-head addition model. The model only uses nodes in token positions P11 to P20. Lower percentages correspond to rarer edge cases. The grey space represents nodes that are not useful.

- The calculation of ST2 in P14.L0.H1 is a interesting case. The model needs ST2 for A2 accuracy. The model could simply reuse the accurate ST2 value calculated in P10. Activation patching shows that it does not. Instead the P14 attention heads calculate ST1 from D1 and D'1 directly, and only relies on the P10.D1.ST2 value in the case where ST2 != ST1. That is, the calculation is "use P14.ST1 value else use ST2 value". This aligns with the model learning the P10.ST1 calculation early in training (for 90% accuracy) and later learning that P10.ST2 contains additional information it can use to get to >99.999% accuracy.

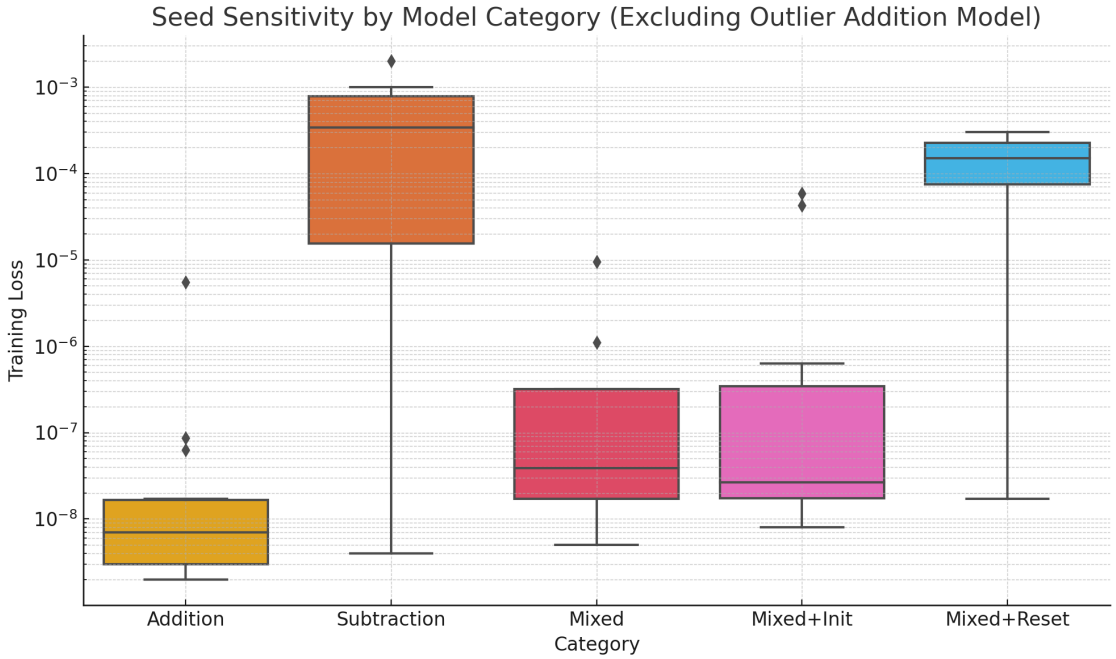

Figure 15: A visualization of the range of training losses across 48 models grouped by the different model categories.

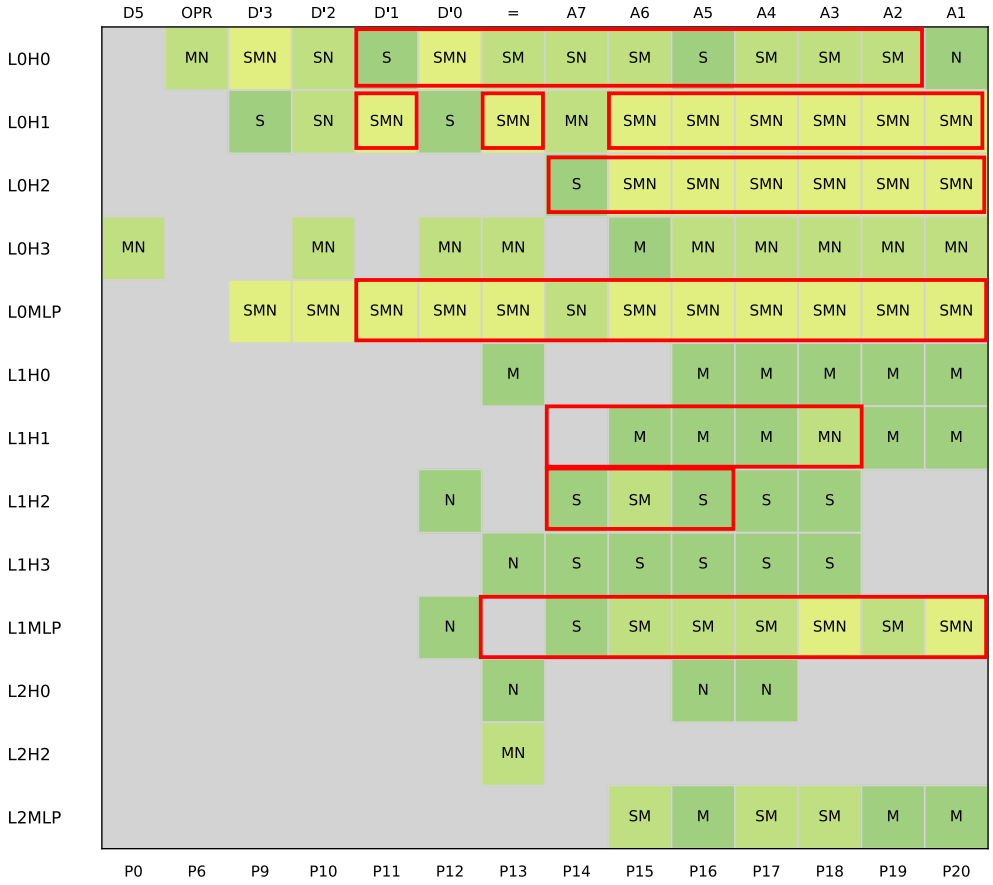

Figure 16: This map of a sample 6-digit **mixed** model shows the 98 nodes used to predict answers to addition (S), positive-answer subtraction (M) and/or negative-answer subtraction (N) questions. Before training the mixed model, 48 nodes were initalized pre-training with a smaller **addition** model's weights. These are have a red border. During mixed model training, 39 of 48 of the initalized monosemantic nodes were generalized (become poly-semantic) and now help predict two or three question classes.